613 Glider ESEKF–SMC: Short Flight-Test Guide

Updated June 19, 2026

1 Purpose and Build Approach

The program is divided into four paths:

1. The IMU is sampled at 80 Hz, while the optional NEO-6M GPS supplies position and velocity at approximately 5 Hz.
2. A 15-state error-state extended Kalman filter (ESEKF) estimates attitude, velocity, position, gyro bias, and accelerometer bias.
3. A soft boundary-layer sliding mode controller (SMC) commands small differential and symmetric wing deflections, with optional tail pitch allocation left disabled until measured.
4. Flight records are placed in a non-blocking queue and written to the SD card by a separate logger thread, so SD latency does not directly stop the control loop.

The first closed-loop flights use a fixed diagonal control-effectiveness matrix B. Online normalized LMS adaptation is intentionally disabled until the fixed-B configuration has been validated. The current control philosophy is deliberately conservative: the aircraft is allowed to keep its natural glide inside a small deadband, and SMC only intervenes when roll, pitch, or angular rate moves outside that band.

2 Current Airframe Assumption

The current MMAV v3 layout has four servo channels: left wing, right wing, D5 morphing deployment, and D6 tail trim. The morphing servo deploys the printed mechanism after startup and is not part of the feedback controller. The tail is also fixed at trim by default; active tail pitch control is present only as a disabled software option because the tail sign and effectiveness have not yet been measured.

The GPS module is treated as optional. It supports ESEKF correction and flight logging, but it is not used as a landing-target controller. A glider without propulsion cannot overcome strong headwind by command alone, and reliable target guidance would require at least a wind or airspeed estimate.

3 ESEKF Summary

The nominal state is

$$\mathbf{x} = [\mathbf{q},\, \mathbf{v},\, \mathbf{p},\, \mathbf{b}_g,\, \mathbf{b}_a]\,,$$

and the 15-dimensional error state is

$$\delta\mathbf{x} = [\delta\boldsymbol{\theta},\, \delta\mathbf{v},\, \delta\mathbf{p},\, \delta\mathbf{b}_g,\, \delta\mathbf{b}_a]^T\,.$$

The gyroscope propagates the quaternion, and the accelerometer propagates velocity and position after gravity compensation. Accelerometer-derived roll and pitch are used only when the measured acceleration magnitude is near g. GPS data, when available, corrects velocity and position; absence of GPS does not stop the filter. Because no compass is installed, yaw is not strongly observable.

4 SMC and Control Effectiveness

For each controlled axis, the sliding surface is

$$s = \dot{e} + \lambda e,$$

where e is the attitude error. The requested angular acceleration is

$$\dot{\boldsymbol{\omega}}_{\mathrm{cmd}} = -\Lambda\boldsymbol{\omega} - K\ \mathrm{sat}\left(\frac{\mathbf{s}}{\boldsymbol{\phi}}\right) - \hat{\mathbf{d}}.$$

The wing commands are calculated from

$$\begin{bmatrix} \dot{p} \\ \dot{q} \end{bmatrix} = \underbrace{\begin{bmatrix} B_{00} & B_{01} \\ B_{10} & B_{11} \end{bmatrix}}_{\mathbf{B}} \begin{bmatrix} u_{\mathrm{diff}} \\ u_{\mathrm{sym}} \end{bmatrix} + \mathbf{d}.$$

For the first flight,

$$\mathbf{B} = \begin{bmatrix} B_{\mathrm{roll}} & 0 \\ 0 & B_{\mathrm{pitch}} \end{bmatrix},$$

with `ENABLE_ADAPTIVE_B=0`. The two diagonal values must be replaced by measured values before closed-loop flight.

5 Physical Quantities

Symbol	Unit	Meaning
$\mathbf{q}$	–	Body-to-world attitude quaternion
ϕ, θ, ψ	rad	Roll, pitch, and yaw angles
p, q, r	rad/s	Body roll, pitch, and yaw rates
$\mathbf{v}$	m/s	World-frame north, east, up velocity
$\mathbf{p}$	m	World-frame north, east, up position
$\mathbf{b}_g$	rad/s	Estimated gyroscope bias
$\mathbf{b}_a$	m/s^2	Estimated accelerometer bias
$\delta\boldsymbol{\theta}$	rad	Small ESEKF attitude error
$\mathbf{P}$	mixed	ESEKF error covariance matrix
λ	s^{-1}	Sliding-surface attitude convergence coefficient
K	rad/s^2	SMC switching acceleration gain
ϕ_{SMC}	rad/s	Boundary-layer width used to reduce chatter
$s_{\mathrm{roll}}, s_{\mathrm{pitch}}$	rad/s	Roll and pitch sliding-surface values
u_{diff}	rad	Differential wing command for roll
u_{sym}	rad	Symmetric wing command for pitch
u_{tail}	rad	Optional tail pitch command, logged but disabled by default
B_{ij}	s^{-2}	Angular-acceleration response per radian of wing command
$\hat{\mathbf{d}}$	rad/s^2	Estimated matched angular-acceleration disturbance
g	m/s^2	Standard gravitational acceleration
Δt	s	Measured control-loop interval

6 Values That Are Not Yet Aircraft Measurements

Code parameter	Current value	Status and reason
B_ROLL_INITIAL	$-6\ \mathrm{s}^{-2}$	Assumed sign and magnitude; replace with measured roll effectiveness.
B_PITCH_INITIAL	$+6\ \mathrm{s}^{-2}$	Assumed sign and magnitude; replace with measured pitch effectiveness.
B_TAIL_PITCH_INITIAL	$+6\ \mathrm{s}^{-2}$	Placeholder only; measure before enabling tail pitch control.
PITCH_TARGET_RAD	0 rad	Placeholder; replace with the measured safe glide trim attitude.
SMC_LAMBDA_*	$0.8\ \mathrm{s}^{-1}$	Soft recovery rate chosen after neutral-wing tests showed over-control risk.
SMC_K_*	$0.6\ \mathrm{rad/s}^2$	Low robustness gain for first controlled tests; increase only after signs are verified.
SMC_PHI_*	0.35 rad/s	Wider boundary layer to reduce wing chatter.
SMC_*DEADBAND*	$5^\circ, 3^\circ, 5^\circ/s$	Allows natural spiral/glide motion before SMC intervenes.
GYRO_NOISE_RADPS	0.015 rad/s	Engineering estimate; refine from stationary IMU logs.
ACCEL_NOISE_MPS2	$0.18\ \mathrm{m/s}^2$	Engineering estimate; refine from stationary IMU logs.
ACCEL_LEVEL_NOISE_RAD	3°	Conservative tilt-observation uncertainty.
SERVO_CMD_LIMIT_DEG	5°	Matches the safer low-authority range suggested by the 0615 flight tests.

7 Software Safety Configuration

The current intended first-flight configuration is:

- `BENCH_MODE=0`: timed automatic arming is disabled.
- `ENABLE_CONTROL=1`: SMC is permitted only after hardware arming.
- `ENABLE_ADAPTIVE_B=0`: B remains fixed and diagonal.
- `USE_TAIL_PITCH_CONTROL=0`: D6 is fixed at tail trim; set to 1 only after measuring tail sign and effectiveness.
- D2 uses `INPUT_PULLUP`; it must first remain open/high and then be connected to ground for at least four control cycles to arm.
- Opening the D2 circuit for at least four cycles disarms the controller, returns both servos to neutral, and clears the controller and B-estimator histories.
- An IMU or ESEKF hard fault is latched and requires a power cycle.

For estimator-only recording, set `ENABLE_CONTROL=0`; IMU, ESEKF, GPS, timing measurements, and SD logging continue while the servos remain neutral.

8 Calibration and Flight-Test Procedure

1. Servo direction and range

Keep the aircraft restrained. Verify neutral PWM, differential motion, symmetric motion, mechanical clearance, and the sign of the resulting roll and pitch commands. Correct `SERVO_LEFT_REVERSE` and `SERVO_RIGHT_REVERSE` before proceeding.

2. IMU axes

Verify that nose-up motion produces increasing pitch, right-wing-down motion produces increasing roll, and the stationary level attitude is close to zero after installation-offset correction. Accelerometer and gyroscope mappings must use the same body-axis convention.

3. Timing and SD test

Run the complete system with SD logging, GPS traffic, and servo activity. Require

$$\texttt{missed\_delta} = 0, \qquad \texttt{control\_hz} \approx 80,$$

and keep `exec_max_us` comfortably below 12500 μs, preferably below approximately 9000 μs.

4. Passive ESEKF test

Set `ENABLE_CONTROL=0`. Record stationary motion, hand rotations, and a low-risk unpowered glide. Check for smooth roll and pitch, reasonable gyro biases, no unexpected ESEKF resets, and no large attitude correction caused by launch acceleration.

5. Measure the diagonal B values

Apply separate, small open-loop differential and symmetric command pulses and estimate

$$B_{\text{roll}} \approx \frac{\Delta \dot{p}}{u_{\text{diff}}}, \qquad B_{\text{pitch}} \approx \frac{\Delta \dot{q}}{u_{\text{sym}}}.$$

A fully locked aircraft without airflow cannot provide aerodynamic B. Use a free-axis rig with airflow or carefully controlled low-energy glide pulses. Repeat the measurement and use a conservative average with the verified signs.

6. Determine pitch trim

Use passive glide data to determine the safe trimmed pitch attitude and replace `PITCH_TARGET_RAD`. Do not assume that geometric zero pitch is the aerodynamic trim attitude.

7. First closed-loop test

Use the measured diagonal B, keep `ENABLE_ADAPTIVE_B=0`, use a reduced physical command range if possible, and perform a short recoverable test. Tune λ, K, and ϕ_{SMC} only after confirming the control signs and estimator behavior.

8. Adaptive-B test

Enable `ENABLE_ADAPTIVE_B=1` only after repeated stable fixed-B flights. Compare the estimated matrix, prediction residuals, commands, and angular accelerations; disable adaptation if cross terms or diagonal magnitudes drift without repeatable excitation.

9 Flight Release Checklist

- Servo directions, neutral positions, linkage limits, and power supply verified.
- IMU axes and installation offsets verified.
- Hardware arm and disarm behavior verified using D2.
- Passive ESEKF flight log reviewed with zero unexplained resets.
- B_{roll}, B_{pitch}, and pitch trim replaced by measured values.
- Fixed diagonal B selected and adaptive B disabled.
- Control timing passes with SD and GPS active.
- First test area, recovery method, and immediate disarm procedure prepared.

10 Update Log

June 15, 2026

- **Update 01:** Replaced the complementary-filter and PD concept with an 80 Hz 15-state ESEKF and boundary-layer SMC while retaining the non-blocking SD logging architecture derived from program 69.
- **Update 02:** Added optional four-pin u-blox NEO-6M support using UBX NAV-POSLLH, NAV-SOL, and NAV-VELNED messages, with absent or invalid GPS data represented by empty log fields instead of runtime errors.
- **Update 03:** Added measured-frequency, maximum-execution-time, missed-period, dropped-log, GPS, and ESEKF-reset telemetry so SD and estimator timing can be evaluated directly on the hardware.

June 17, 2026

- **Update 04:** Corrected angle wrapping, increased GPS local-coordinate conversion precision, corrected covariance initialization order, refreshed Euler angles after estimator reset, and removed a redundant Euler-angle extraction.
- **Update 05:** Added an initial arming delay, commanded neutral servos after an IMU read failure, and introduced a 16-cycle ESEKF settling interval before control could resume after a reset.
- **Update 06:** Reset the B-estimator derivative history after an IMU interruption so an old angular-rate sample could not create a false angular-acceleration impulse when IMU data returned.
- **Update 07:** Added gyroscope-calibration variance checking so vibration or changing angular motion during startup calibration causes calibration rejection instead of being stored as sensor bias.
- **Update 08:** Replaced the absolute B-matrix determinant threshold with a threshold relative to the diagonal product to reduce unsafe inversion of a nearly singular effectiveness matrix.
- **Update 09:** Required the B-matrix determinant to have the expected negative sign so excessive cross-axis estimates cannot pass the inverse-matrix check merely because their determinant magnitude is large.
- **Update 10:** Added acceleration-magnitude and mean-rate checks to startup calibration, tightened the acceptable gyro mean, and reduced the IMU read timeout from 20 ms to 8 ms to fit within the 12.5 ms control period.
- **Update 11:** Added separate bench/flight arming paths, a D2 hardware arm input, an IMU fault latch, an ESEKF fault latch, and a compile-time option that disables adaptive B estimation for initial flights.
- **Update 12:** Reordered the loop so IMU acquisition, ESEKF propagation, GPS correction, timing measurement, and SD logging continue while disarmed, preventing stale attitude from being used at the moment of arming.
- **Update 13:** Added `ENABLE_CONTROL` as an independent estimator-only mode so passive ESEKF/GPS/SD testing keeps both servos neutral without disabling sensor processing or logging.
- **Update 14:** Changed the default build to hardware arming, required the D2 input to be observed released before it can arm, and made removal of the arm signal actively disarm the controller and return both servos to neutral.
- **Update 15:** Added a shared control-state reset function that clears filtered commands, servo state, disturbance estimates, angular-acceleration history, and B-estimator readiness whenever control is armed, disarmed, or faulted.
- **Update 16:** Added four consecutive control-cycle confirmation for both D2 arm and disarm transitions so vibration or a one-frame electrical disturbance cannot momentarily reset and immediately restart the controller.
- **Update 17:** Unified IMU and ESEKF fault cleanup through the shared reset function, eliminating the earlier incomplete manual cleanup that left matched-disturbance state unchanged.
- **Update 18:** Reset the complete B matrix to the configured initial diagonal values on every control-state reset so a poor adaptive estimate cannot survive a disarm and contaminate the next arming cycle.
- **Update 19:** Cleared s_{roll} and s_{pitch} during control-state reset so disarmed and faulted log records no longer retain stale sliding-surface values.

June 18, 2026

- **Update 20:** Added a third D5 morphing servo that starts at the stowed angle, deploys to the configured morphing angle after one second, and logs its commanded angle and PWM without entering the SMC loop.
- **Update 21:** Matched the 0615 tested code style by adding a per-flight log tag, Core 0/Core 1 SD status messages, CSV stop markers, calibration output, per-side wing neutral/gain/incidence-trim servo mapping, and applied-incidence log columns.
- **Update 22:** Changed the D5 morphing servo deployment from 130° to 150°, giving a 60° motion from the 90° stowed position.
- **Update 23:** Added a fixed D6 tail-trim servo that is commanded to the configured neutral angle at startup, logged as tail angle and PWM, and left outside the ESEKF and SMC control loops.

June 19, 2026

- **Update 24:** Integrated the 0615 flight-test lesson by changing SMC into a neutral-biased soft stabilizer with small authority, deadbands, launch-delay intervention, explicit tail-trim fallback, optional tail pitch allocation, and added u_{tail} plus fault/armed status telemetry.
- **Update 25:** Reworked this guide to match the current four-servo MMAV v3 assumptions, clarify that GPS is not yet a landing-target controller, and group the update history by date.